

State-Level Social and Economic Resilience After COVID-19

Evaluating post-COVID resilience using ACS state-level economic and social indicators
Data: American Community Survey (ACS) 2011–2024 | 52 states/territories

Manana Asatiani, Adeline Chen & Madison Davis

Research Question & Analytical Framework

ACS state-level data, 2011–2024 | Pre-COVID (2011–2019) vs. Post-COVID (2021–2024)

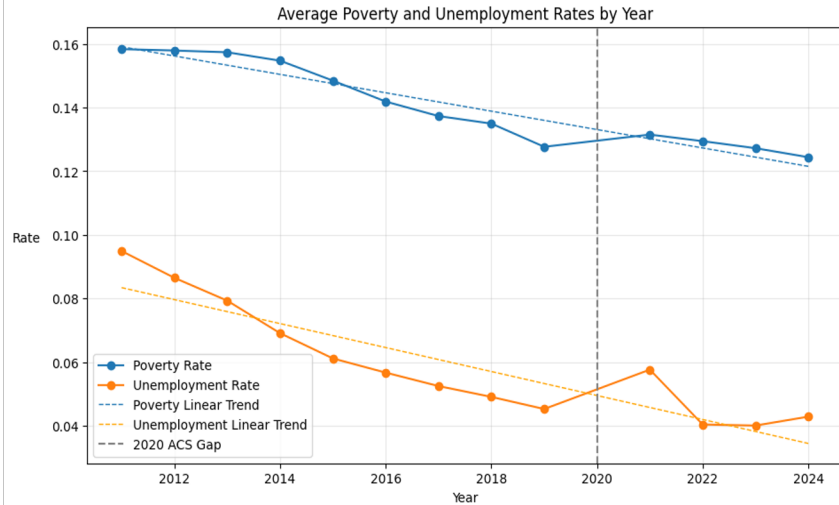
Research Question

How did state-level economic and social resilience vary before and after COVID-19, and which conditions appear most associated with stronger recovery?

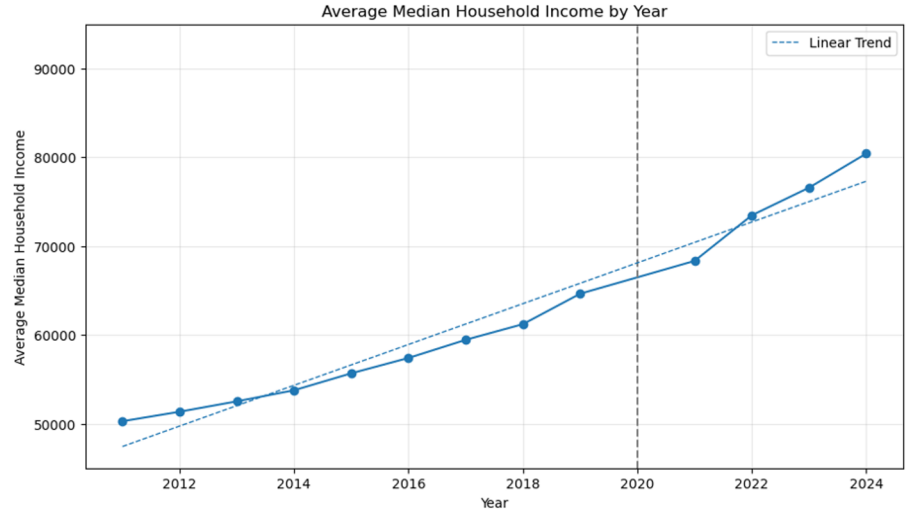
Data & Scope	Indicators Analyzed	Methods
▶ ACS 1-Year Estimates (Census API) ▶ 52 states + DC + Puerto Rico ▶ 13 years: 2011–2019, 2021–2024 ▶ 2020 excluded — no ACS 1-yr release	▶ Income, poverty, unemployment ▶ Education attainment ▶ Housing cost burden, overcrowding ▶ Uninsured rates, remote work, commuting ▶ Relationship data	▶ Feature engineering (rates/ratios) ▶ Pre vs. Post-COVID comparison ▶ Correlation & regression analysis ▶ Composite Social Resilience Index

Economic conditions improved after COVID through rising incomes and lower unemployment, but poverty reductions lagged

National average, 2011–2024 | ACS 1-Year Estimates



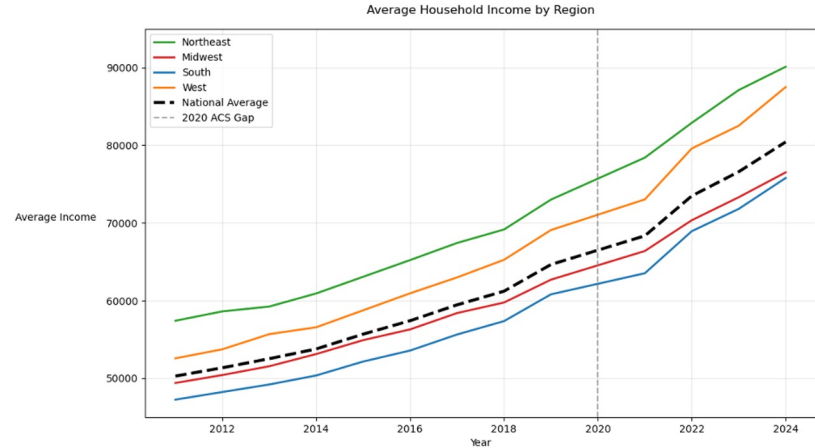
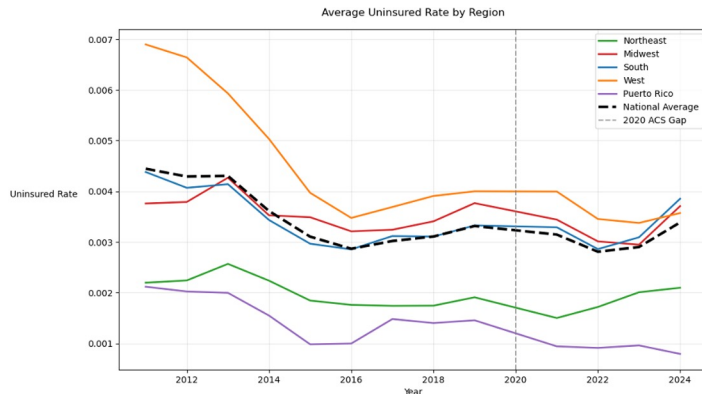
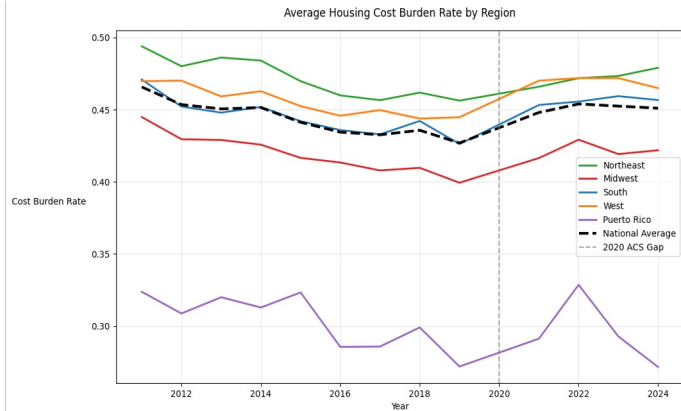
- Unemployment fell from ~9.5% to ~4% signaling strong labor market recovery.
- Poverty rate declined more slowly: 15.9% → 12.4%. COVID did not erase structural poverty.
- Key insight: The labor market bounced back but structural economic vulnerability persisted.



- Avg. income grew from ~\$50K (2011) to ~\$80K (2024). 60% nominal increase.
- Post-COVID acceleration: income grew faster 2021–2024 than any prior 4-year period.
- Income recovery was the strongest headline signal but alone can overstate recovery

Housing Affordability and Insurance Pressures Remained Elevated Even as Incomes Rose

Average housing cost burden rate and uninsured rate by region, 2011–2024



~45% of renters nationally are cost-burdened — paying 30%+ of income on rent. This barely moved post-COVID.

Income recovery and housing affordability are decoupled — a critical insight for policy.

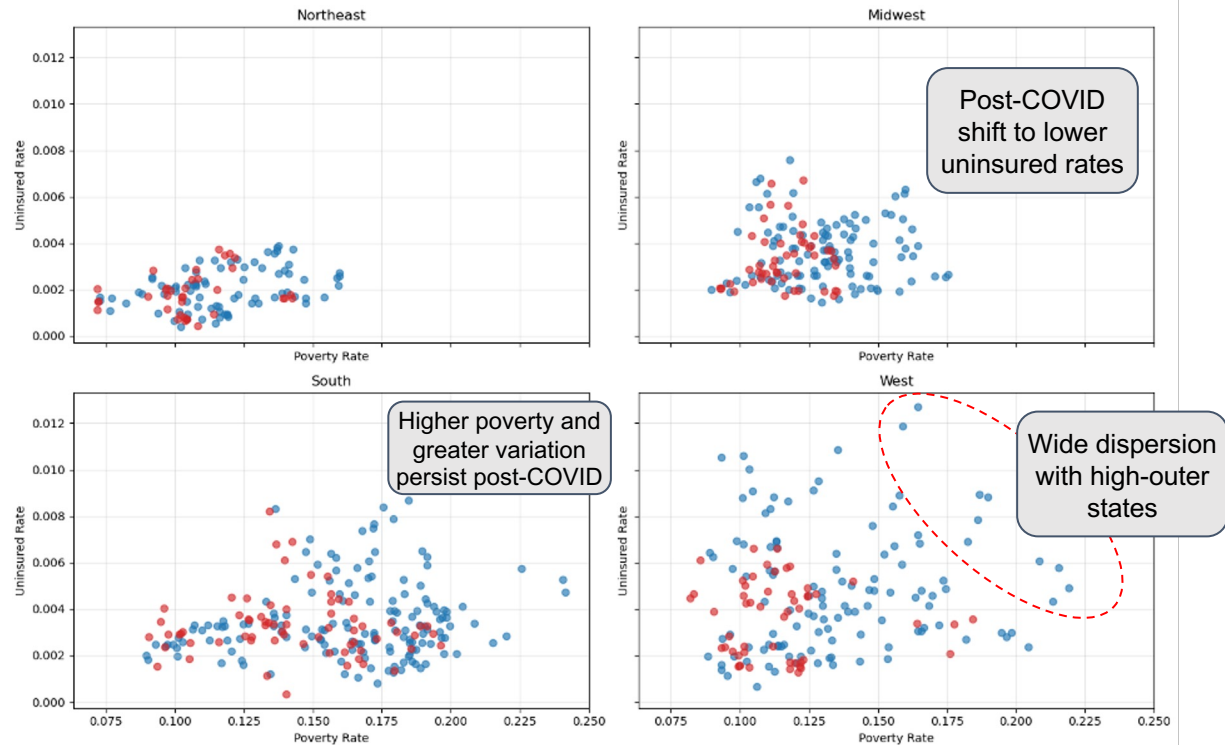
Northeast and West: highest burden despite highest incomes. Stronger wages didn't solve affordability.

Regional disparities in poverty and insurance access persisted after COVID despite modest post-pandemic improvements

Income growth vs. uninsured rate improvement by region | Pre-to-Post COVID

Poverty and Uninsured Rates by Region: Pre vs Post COVID

● Pre-COVID ● Post-COVID



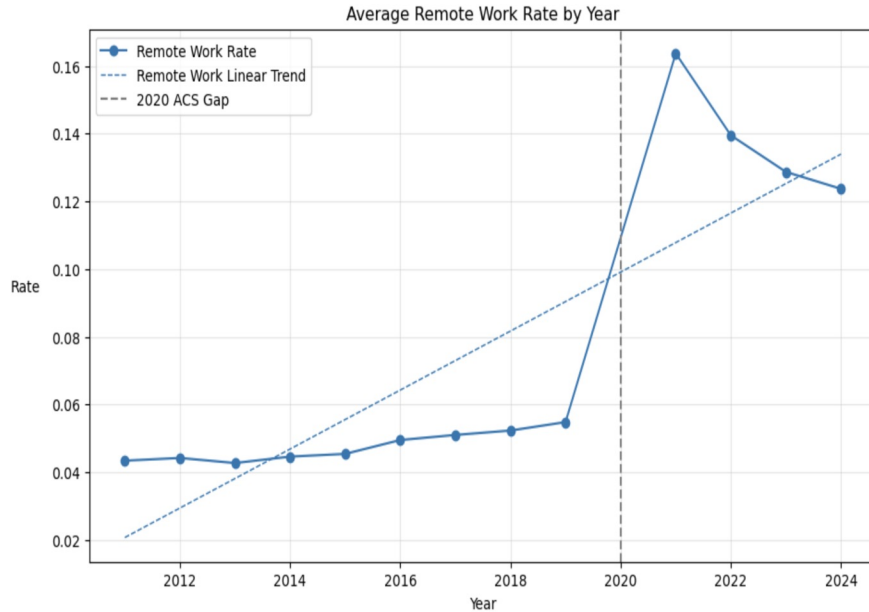
Poverty and uninsured rates remained positively related across all regions. States with higher poverty levels generally experienced higher uninsured rates both before and after COVID.

Post COVID observations shifted slightly downward in several regions. Many states showed slightly lower uninsured rates at similar poverty levels post-COVID. This was especially prominent in the South and Midwest

Regional disparities remained substantial after recovery. The South and the West continued to display wider variation and higher overall poverty rates than the Northeast.

COVID Permanently Restructured How Americans Work and Commute

Remote work rate, national average 2011–2024



Remote Work Rate (2011–2024)

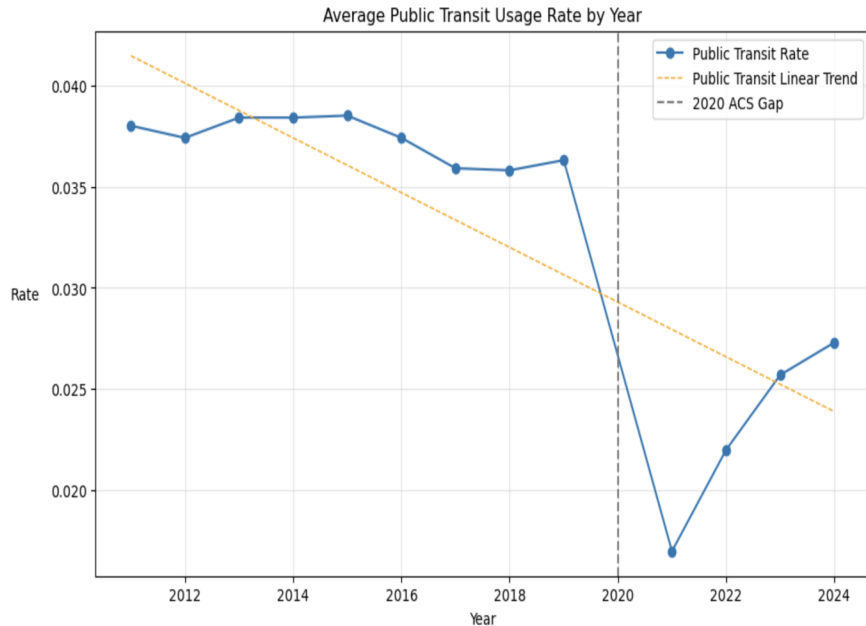
Remote work adoption accelerated significantly after COVID-19

- Remote work remained relatively stable at ~4–6% between 2011–2019
- Adoption increased to over 16% in 2021 following the pandemic
- Remote work rates declined slightly afterward but stabilized well above pre-pandemic levels
- The data suggests COVID-19 accelerated a long-term shift toward hybrid and flexible work arrangements

Remote work: 5% → 16% post-COVID, stabilized ~12% by 2024

COVID Permanently Restructured How Americans Work and Commute

Public transit usage, national average 2011–2024



Public Transit Usage Rate (2011–2024)

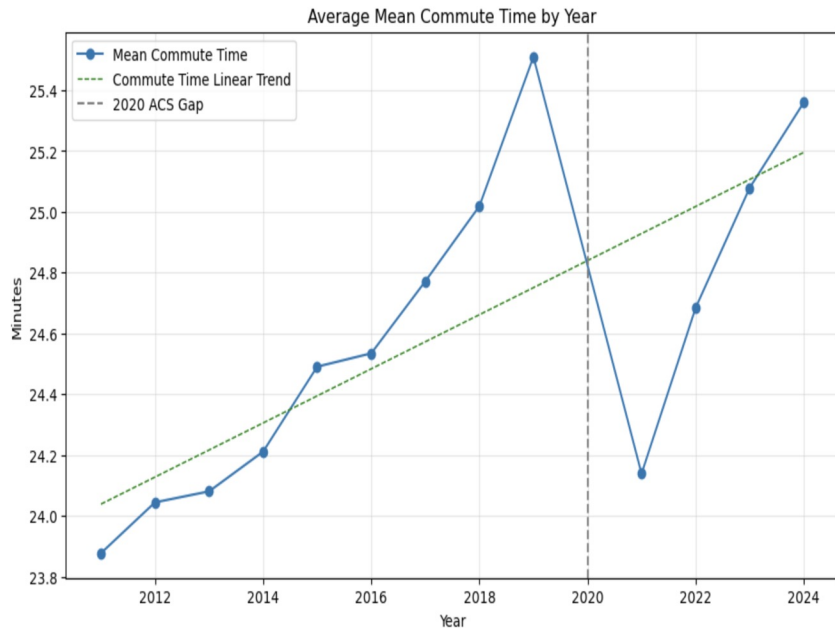
Transit usage: 4% → 1.7% post-COVID, still below pre-pandemic levels by 2024

Public transit usage declined significantly following COVID-19

- Transit usage remained relatively stable at ~3.5–4.0% between 2011–2019
- Average transit usage fell to roughly 1.7% in 2021 following the pandemic
- Transit usage recovered gradually afterward but remained below pre-pandemic levels through 2024
- The decline suggests many workers did not fully return to traditional commuting patterns, partly due to continued remote and hybrid work arrangements

COVID Permanently Restructured How Americans Work and Commute

Mean commute time, national average 2011–2024



Mean Commute Time (2011–2024)

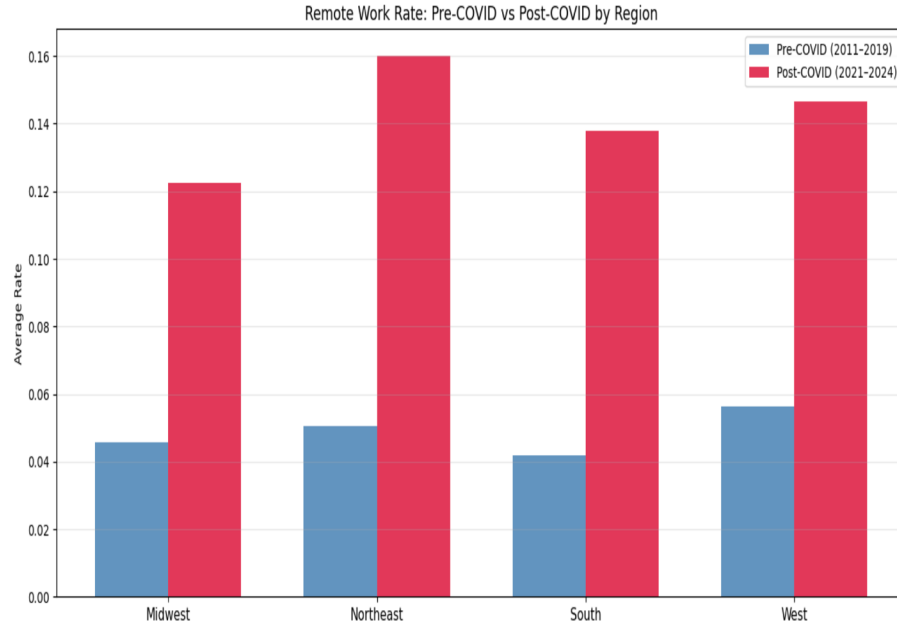
Commute times changed less dramatically following COVID-19

- Commute times gradually increased between 2011–2019, rising from ~24 to 25.5 minutes
- Average commute times declined in 2021, likely reflecting increased remote and hybrid work
- Commute times gradually increased again between 2022–2024, returning close to pre-pandemic levels
- Unlike remote work and transit usage, commute patterns recovered more gradually toward long-term historical trends

Commute times briefly declined post-COVID before returning close to pre-pandemic levels by 2024

COVID Permanently Restructured How Americans Work and Commute

Remote work rate, national average 2011–2024



Remote Work Rate (2011–2024)

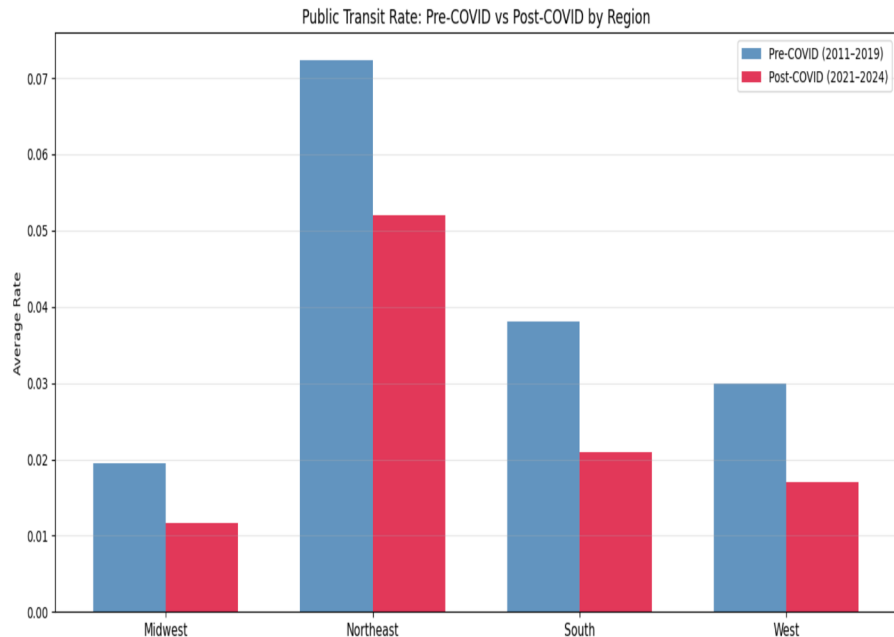
Remote work adoption varied across U.S. regions

- **All regions** experienced large increases in remote work adoption following COVID-19
- The **Northeast** experienced the largest increase, rising from 5% to 16% post-pandemic
- The **West** had the second-highest remote work adoption after COVID-19
- The **Midwest** and **South** remained somewhat lower overall, likely reflecting differences in industry composition and office-based employment

The Northeast and West experienced the strongest long-term shift toward remote and hybrid work

COVID Permanently Restructured How Americans Work and Commute

Public transit usage, national average 2011–2024



Public Transit Usage Rate (2011–2024)

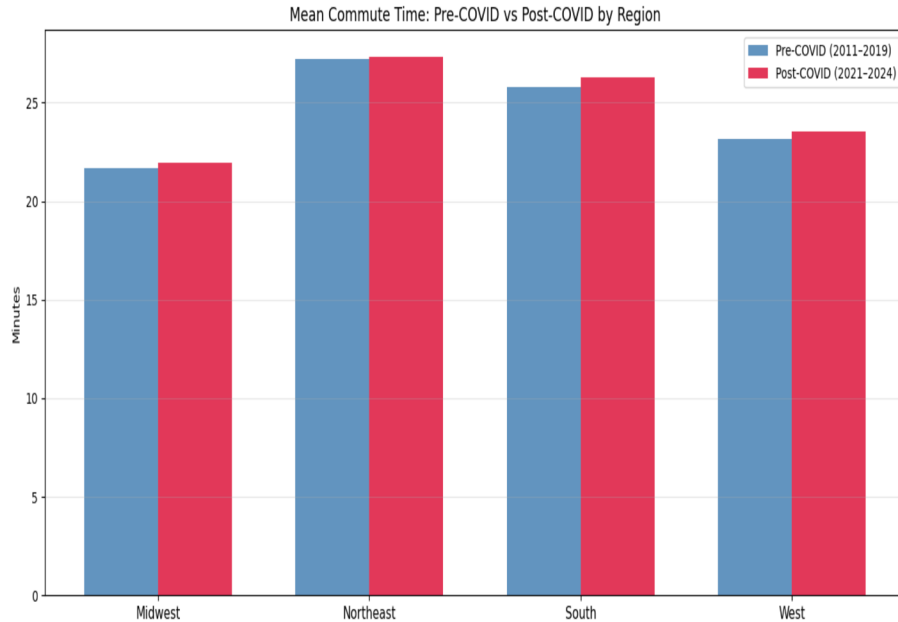
Public transit usage declined across all U.S. regions

- **All regions** experienced declines in public transit usage following COVID-19
- The **Northeast** remained the most transit-dependent region despite large declines post-pandemic
- Transit usage in the South, Midwest, and West remained lower overall
- The broad decline suggests commuting behavior shifted nationwide as remote and hybrid work became more common

Transit usage declined nationwide as fewer workers returned to traditional commuting patterns

COVID Permanently Restructured How Americans Work and Commute

Mean commute time, national average 2011–2024



Mean Commute Time (2011–2024)

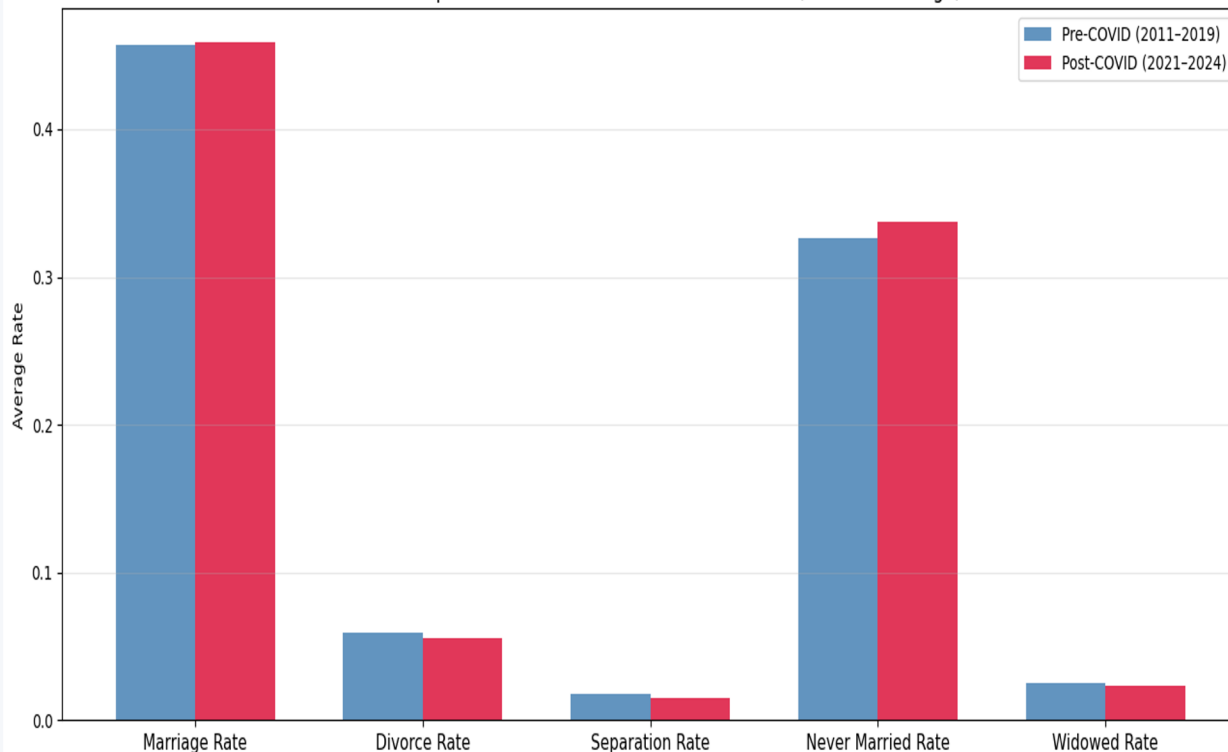
Commute times briefly declined post-COVID before returning close to pre-pandemic levels by 2024

Regional commute times remained relatively stable post-COVID

- **Commute times** changed less dramatically than remote work and transit usage across all regions
- The **Northeast** has the longest average commute times before and after COVID-19
- Most regions experienced only slight increases in commute times post-pandemic
- Longer commute times may reflect people moving farther away from major cities after COVID

The National Picture: Smaller Shifts Than Expected

Relationship Status Rates: Pre-COVID vs Post-COVID (National Average)



45.9%

Married (post)

33.8%

Never Married (post)

5.6%

Divorced (post)

1.5%

Separated (post)

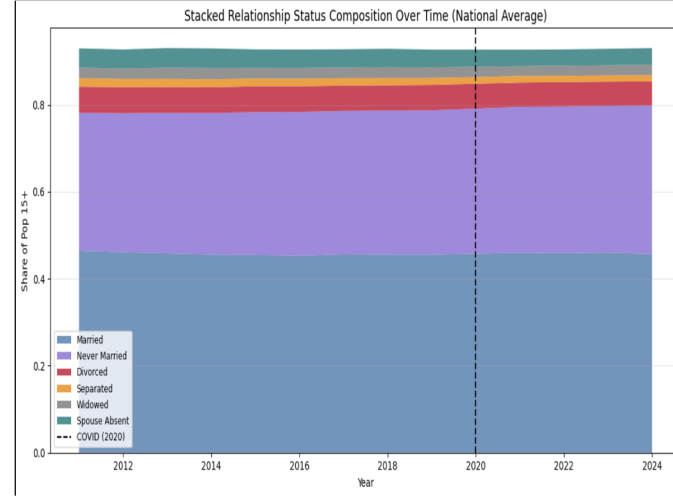
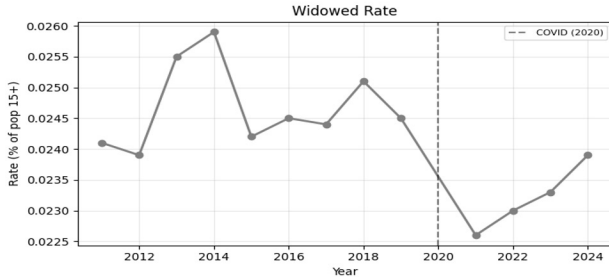
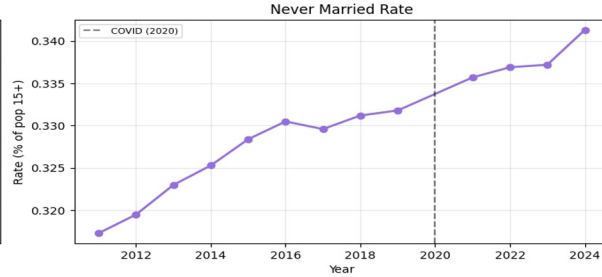
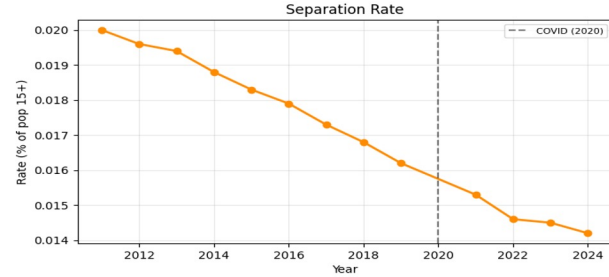
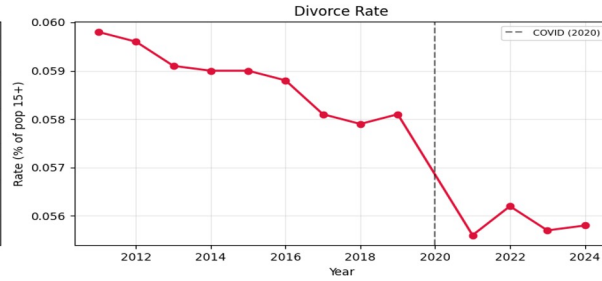
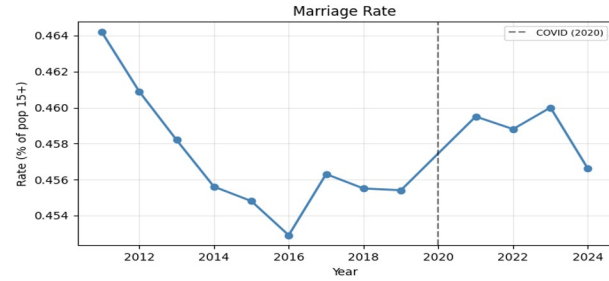
Why this matters

COVID did not cause a sudden collapse in household structure. Changes were modest: marriage nearly flat, formal dissolution fell slightly, and the more meaningful shift was a continued rise in never-married adults.

COVID reinforced long-run demographic trends, not a new crisis.

Social Recovery: Household & Relationship Indicators Tell a Different Story

National Relationship Status Rates, 2011–2024



Divorce & Separation: Declining

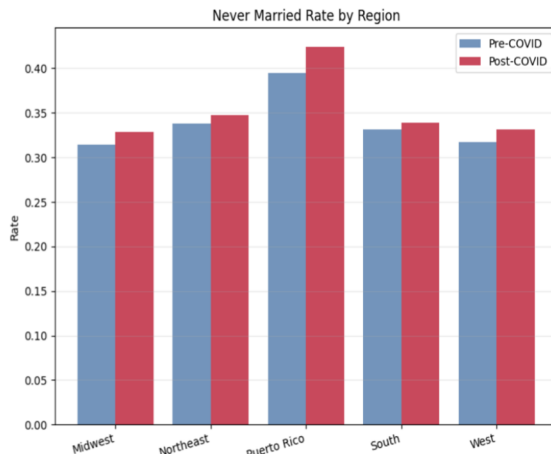
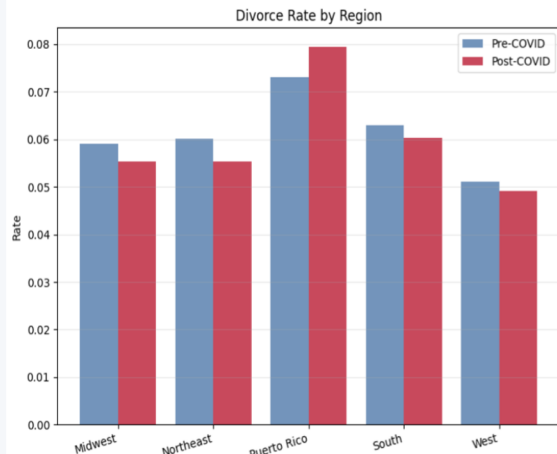
Divorce fell from 5.9% to 5.6%; separation from 1.8% to 1.5%. Lower rates may reflect financial constraints and delayed legal action not necessarily stronger relationships.

Married Rate: Nearly Flat

45.7% → 45.9%. A tiny tick upward. The stacked area chart shows the overall composition has been remarkably stable - household structure changes slowly even under major disruption.

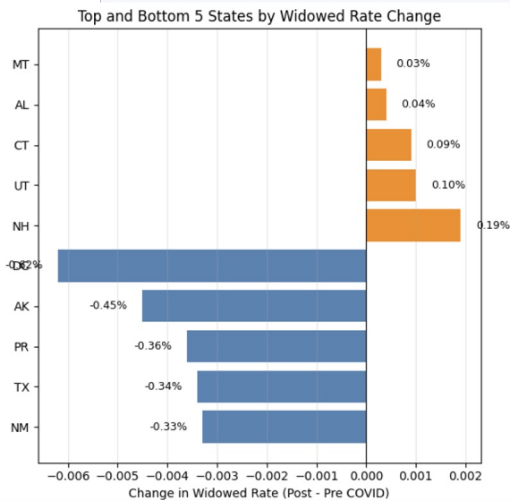
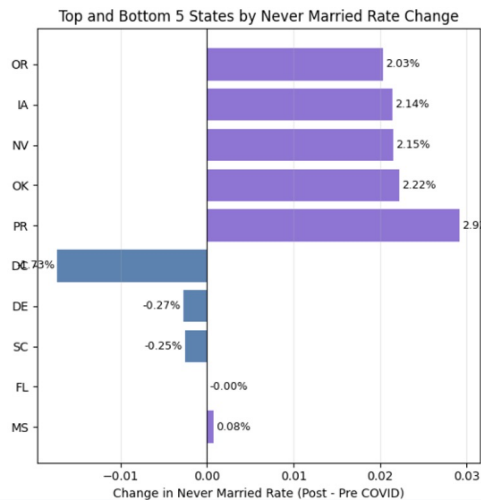
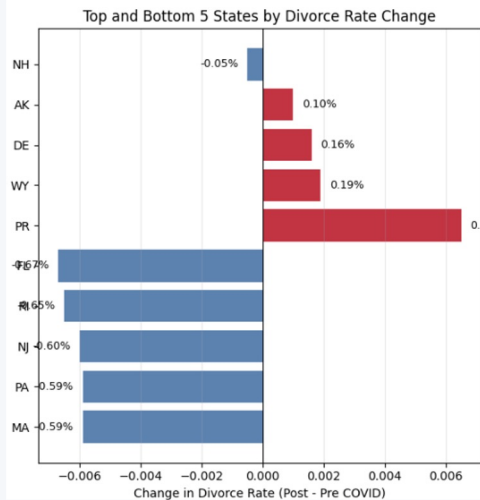
Regional Variation: Geography Shaped the Shift

Regional Relationship Rates: Pre vs Post COVID



Puerto Rico stands out

Highest divorce rates and the steepest rise in never-married adults post-COVID — likely reflecting economic precarity compounding pre-existing household fragility.

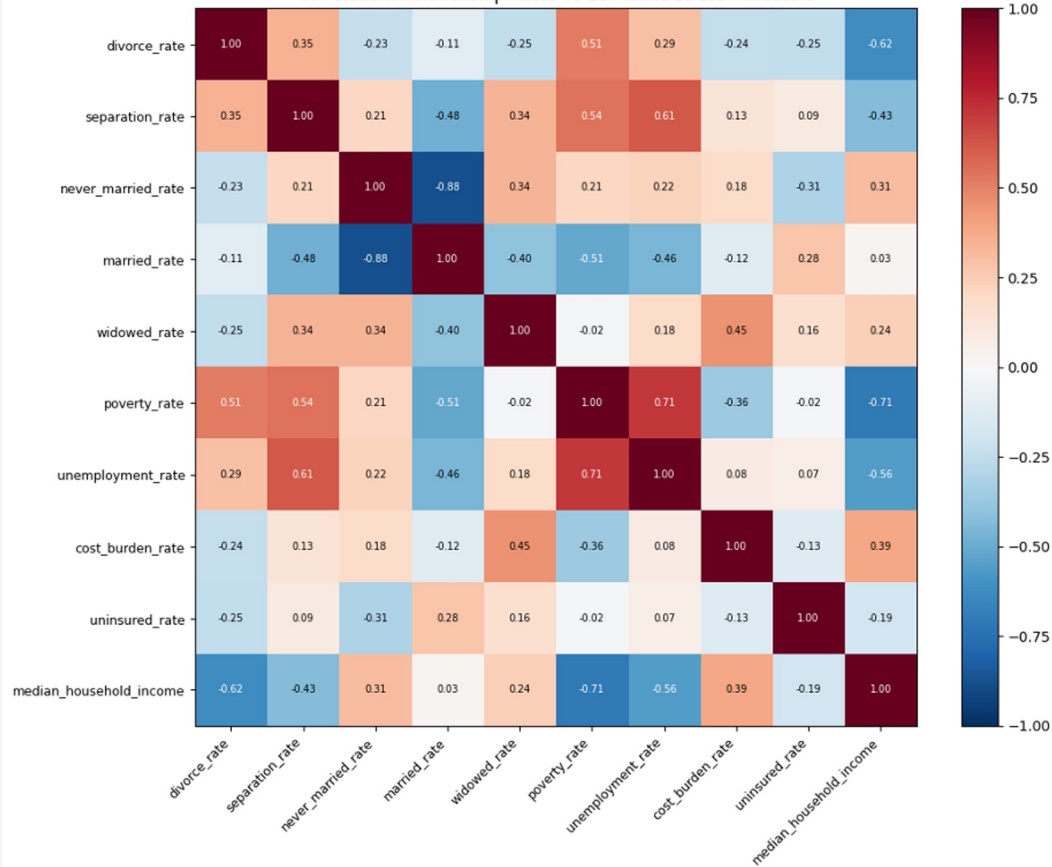


Northeast led divorce decline

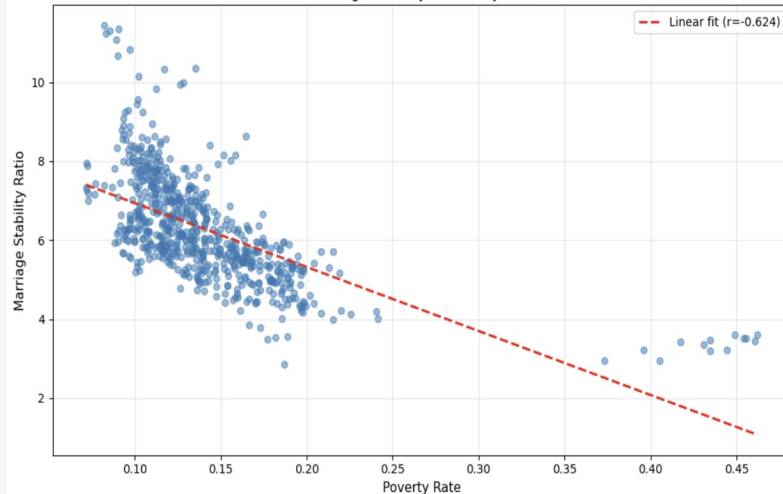
MA, NJ, PA, and RI showed the largest drops in divorce rates. Higher incomes and urbanization may have insulated these states from household-dissolution pressure.

What Drives Instability? Economic Stress.

Correlation: Relationship Rates vs Economic Stress Indicators



Marriage Stability vs Poverty Rate



Key correlations

Poverty $\leftrightarrow$ Divorce: $r = +0.51$

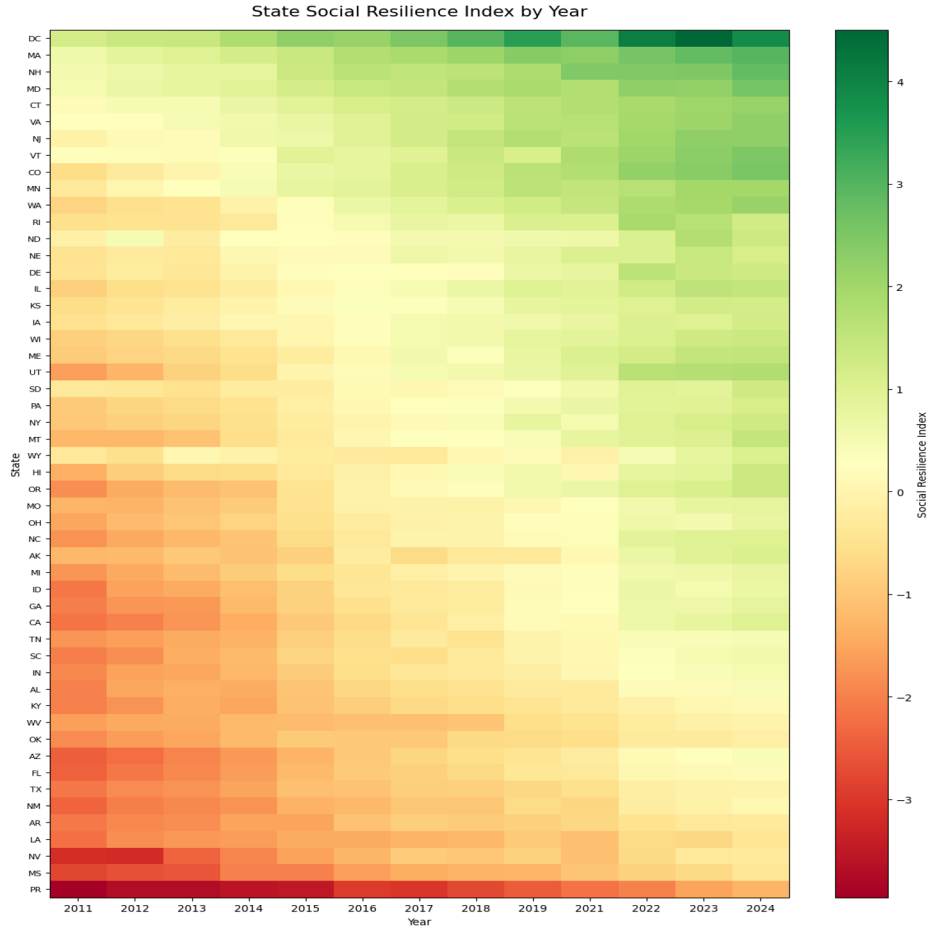
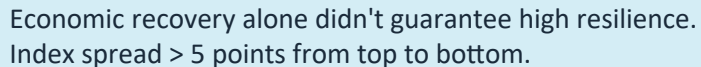
Poverty $\leftrightarrow$ Separation: $r = +0.54$

Income $\leftrightarrow$ Divorce: $r = -0.62$

Marriage $\leftrightarrow$ Never-Married: $r = -0.88$

Higher poverty rates are strongly associated with more divorce and separation. Marriage stability falls sharply as poverty rises ($r = -0.624$ scatter above).

Index combines income, education, poverty, unemployment, uninsured, housing burden, and overcrowding (red = lower resilience, green = higher)

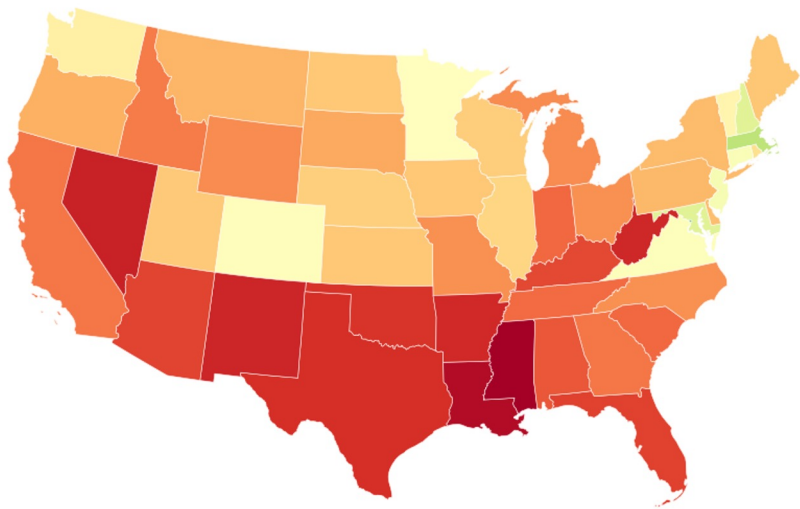


Post-COVID recovery was uneven, with resilience gains more concentrated in select regions

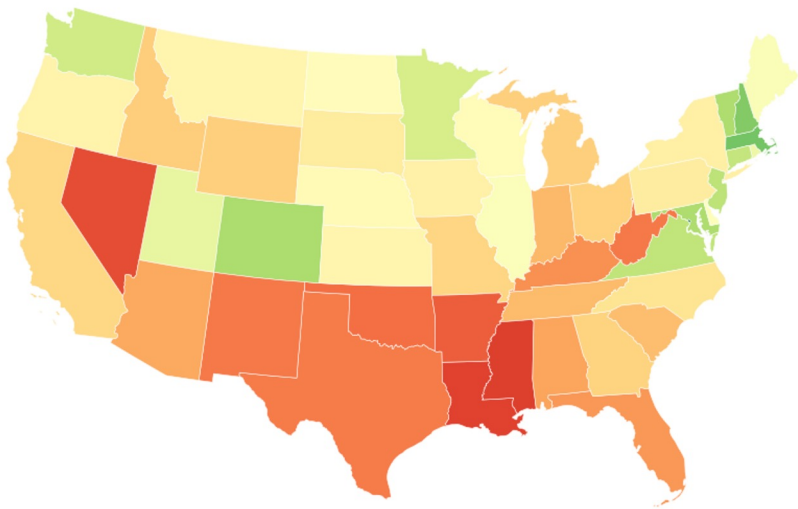
State Social Resilience Index average (red = lower resilience, green = higher)

Social Resilience Index by State: Pre-COVID vs Post-COVID

Pre-COVID Average, 2016-2019



Post-COVID Average, 2021-2024



Key Takeaways

Recovery After COVID Was About More Than Economics

Income & Employment Recovered

Headline economic metrics improved substantially post-COVID but poverty declined more slowly, signaling incomplete financial recovery.

Housing Affordability Remained a Pressure Point

~45% of renters stayed cost-burdened even as incomes rose. Income increased, but rent increased at a faster rate.

Work Patterns Shifted Structurally

Remote work tripled and transit use dropped. These changes have not fully reversed and are reshaping urban economies.

Education Is the Strongest Resilience Driver

Education rate showed the highest correlation with resilience ($r = 0.85$). Long-term educational investment pays off in crisis resilience.

Relationship Structure Did Not Change Significantly

Pre and Post Covid family trends continued for the most part and reflect current social trends in marriage.

Structural Gaps Pre-Dated COVID and Persisted

Bottom-performing states (PR, MS, LA, NV) reflected pre-existing vulnerabilities that COVID did not create but rather amplified.